

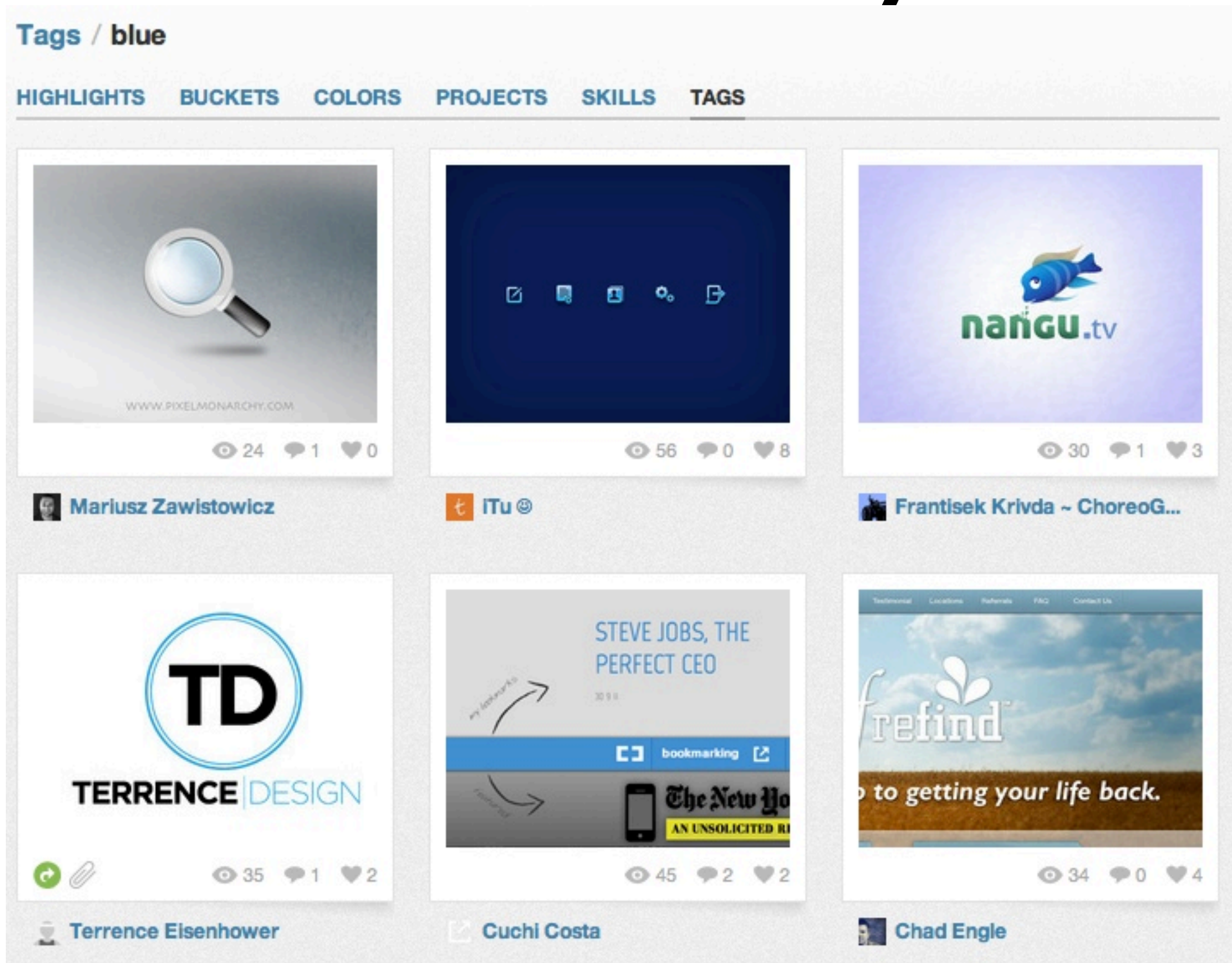
Colors at Dribbble



Historically:



Historically:



Search by Color!



Demo!



Processing

- Colors in shot
- Amount of color in shot

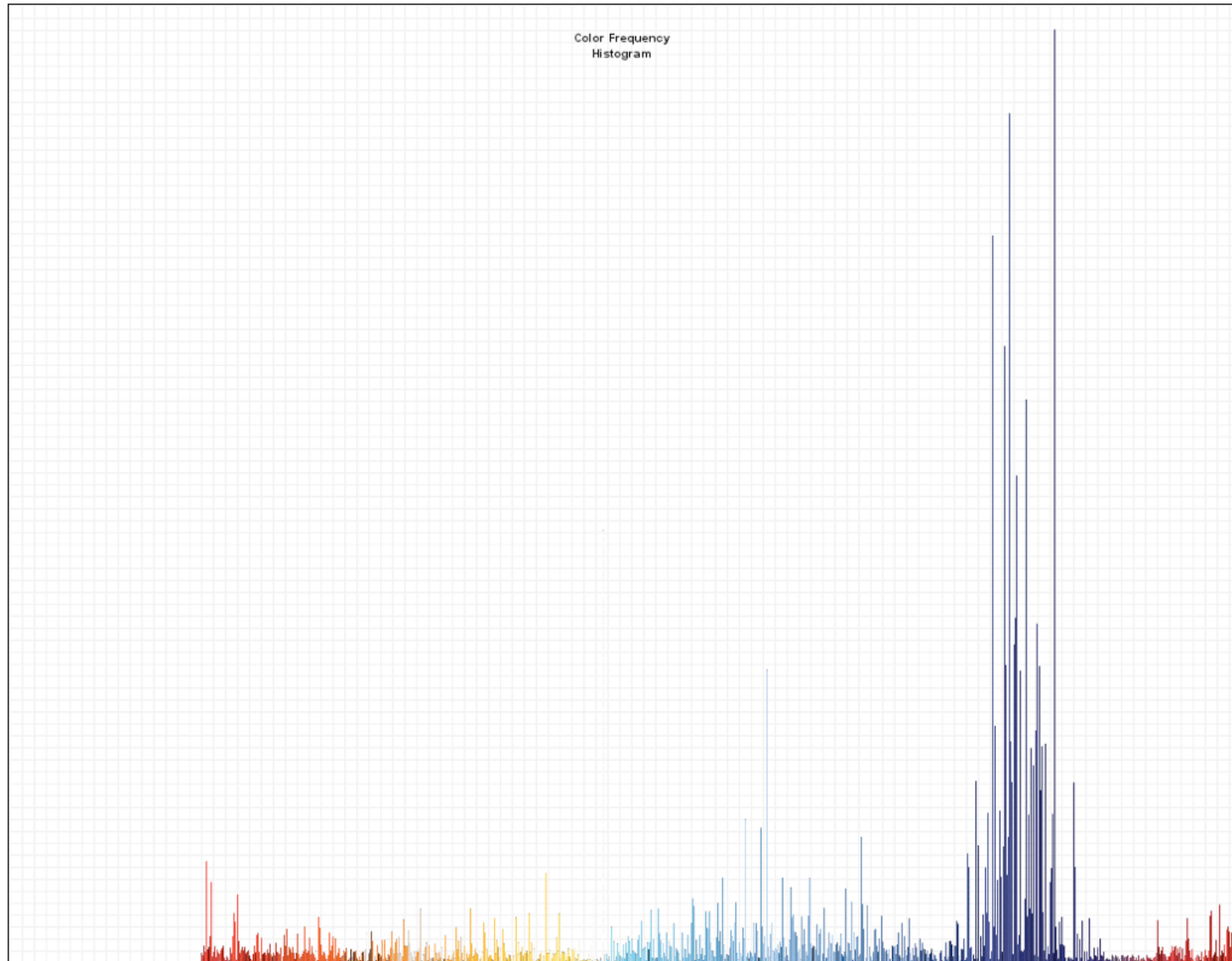


Color Histogram!

- Colors in shot
- Amount of color in shot



Color Histogram



Problem

120,000 Colors Max



Problem

120,000 Colors Max * 200,000 Shots



Problem

120,000 Colors Max * 200,000 Shots
= 24 Billion Rows



Problem



Problem



(23⁴, 76, 136)

(23³, 76, 136)



Problem



(234,76,136)

Quantize it!

- “Round” colors
- Fewer colors
- Preserves accuracy



Quantize it!



(23⁴,76,136)

(23³,76,136)

Quantized!



(234,76,136)



Example



Space Dog — Melanie Matthews



32 Colors



4 Colors

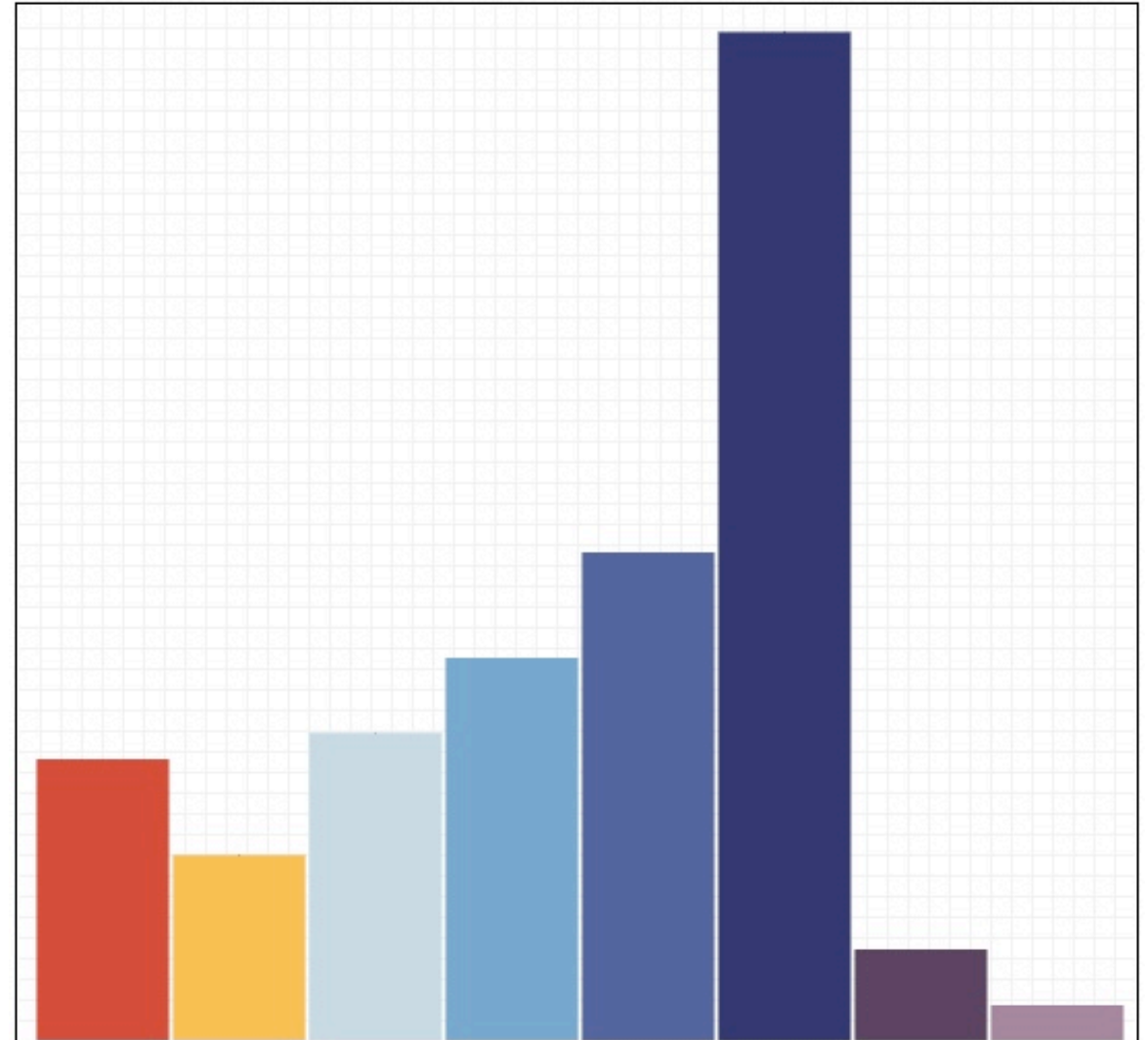




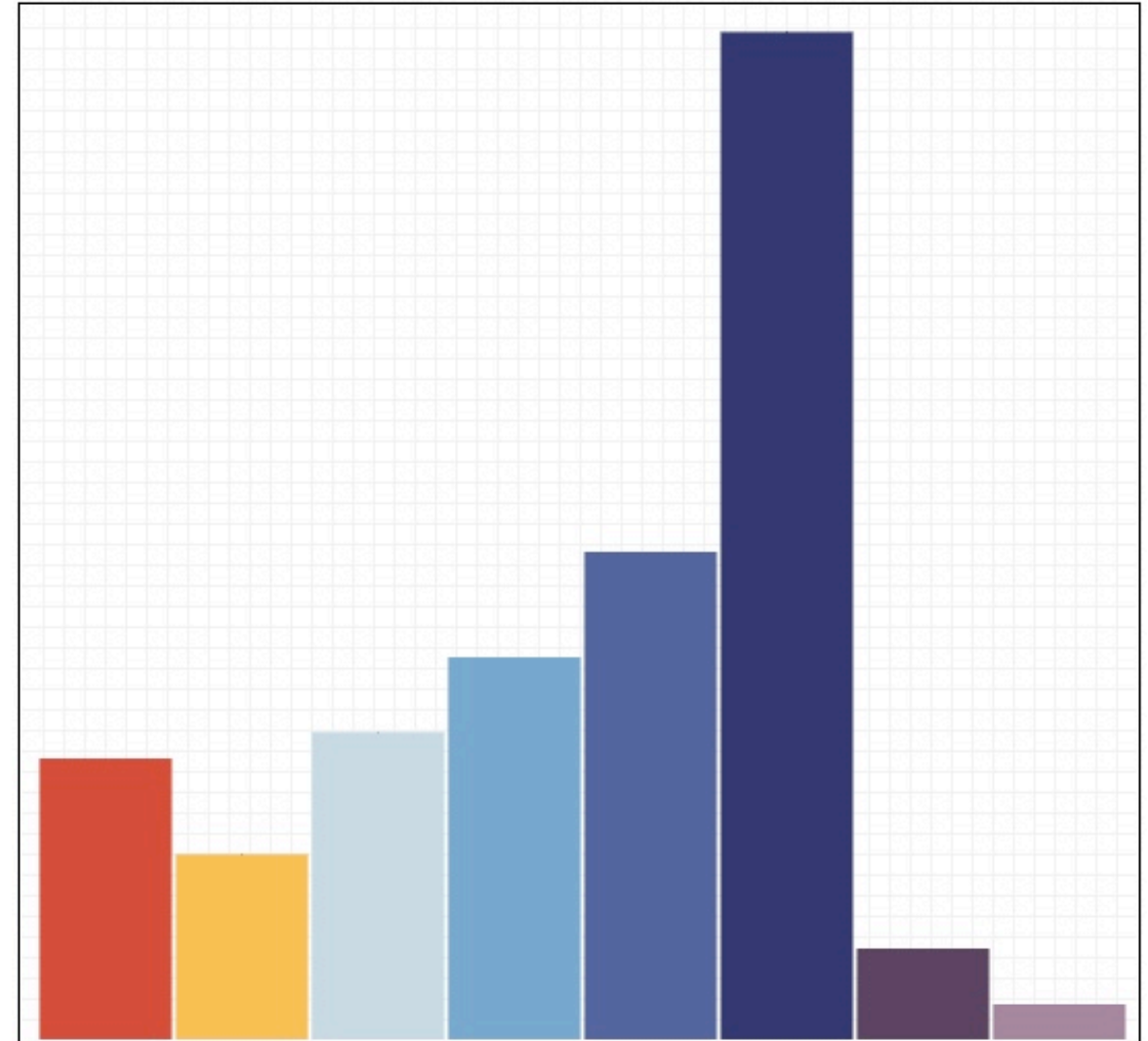
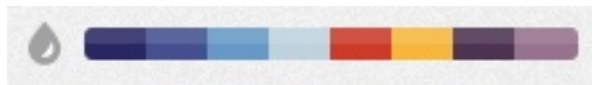
8 Colors



Quantized Histogram



Quantized Histogram



Store in DB

```
dribbble=# select * from colors where screenshot_id = 1;
```

id	screenshot_id	red	green	blue	hex	percentage	created_at
738	1	60909	60652	58596	EDEDE5	0.4198666666666667	2011-05-17 20:28:58.725078
737	1	20560	16962	18504	504249	0.022275	2011-05-17 20:28:58.717163
736	1	13107	13107	21588	343454	0.1521	2011-05-17 20:28:58.711474
735	1	24415	22102	25700	605764	0.048	2011-05-17 20:28:58.707377
734	1	40606	37008	38293	9F9195	0.1343666666666667	2011-05-17 20:28:58.702557
733	1	7196	6425	8224	1C1A20	0.09258333333333333	2011-05-17 20:28:58.697289
732	1	34695	32382	33667	877E83	0.07008333333333333	2011-05-17 20:28:58.689664
731	1	39321	26728	22873	99695A	0.060725	2011-05-17 20:28:58.677548

(8 rows)

Search!



Query

Find shots with the color that you're searching for.



Searching Problem

8 Colors per shot



Searching Problem

8 Colors per shot * 200,000 Shots



Searching Problem

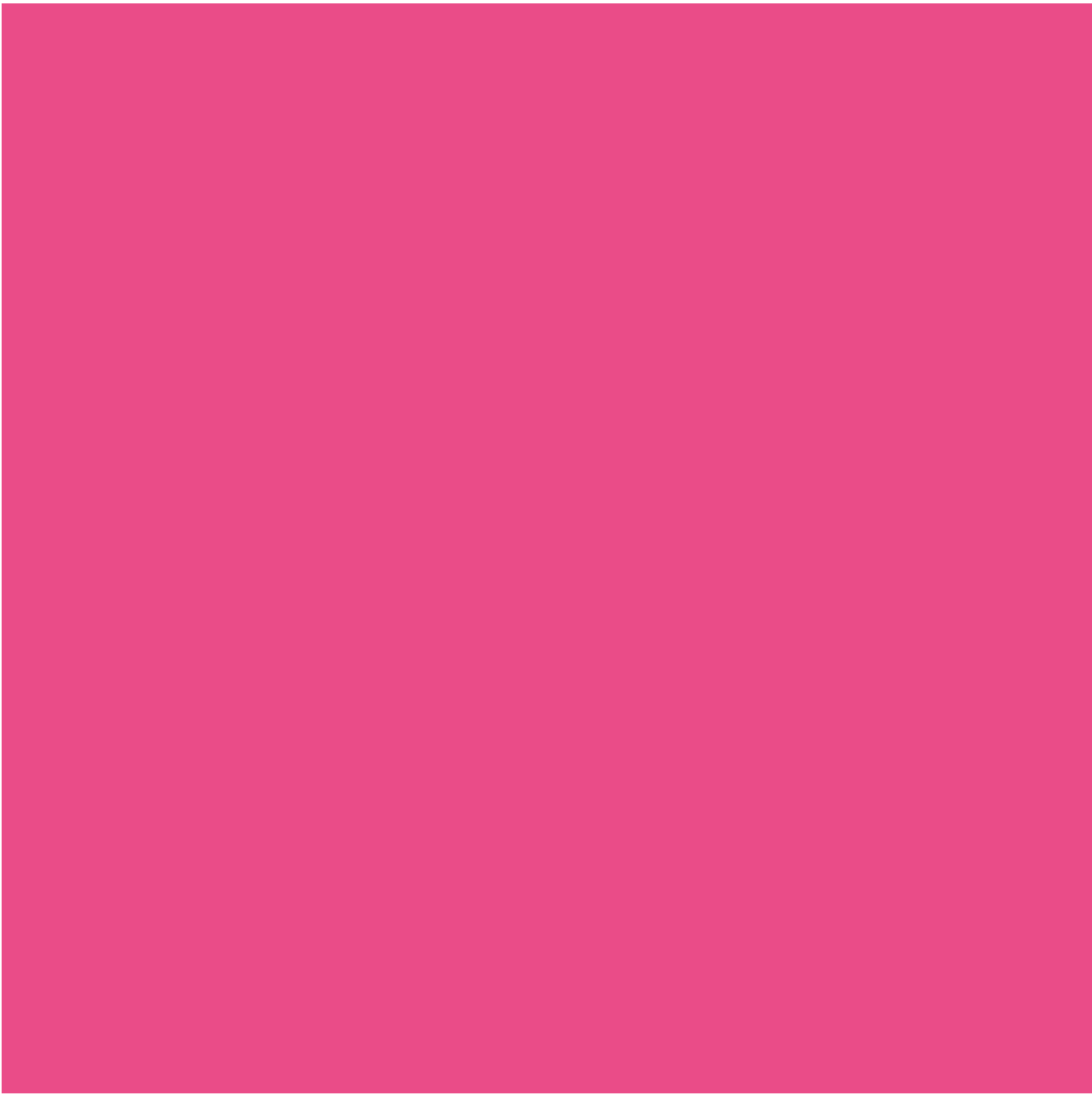
8 Colors per shot * 200,000 Shots
= 1.6 Million Colors



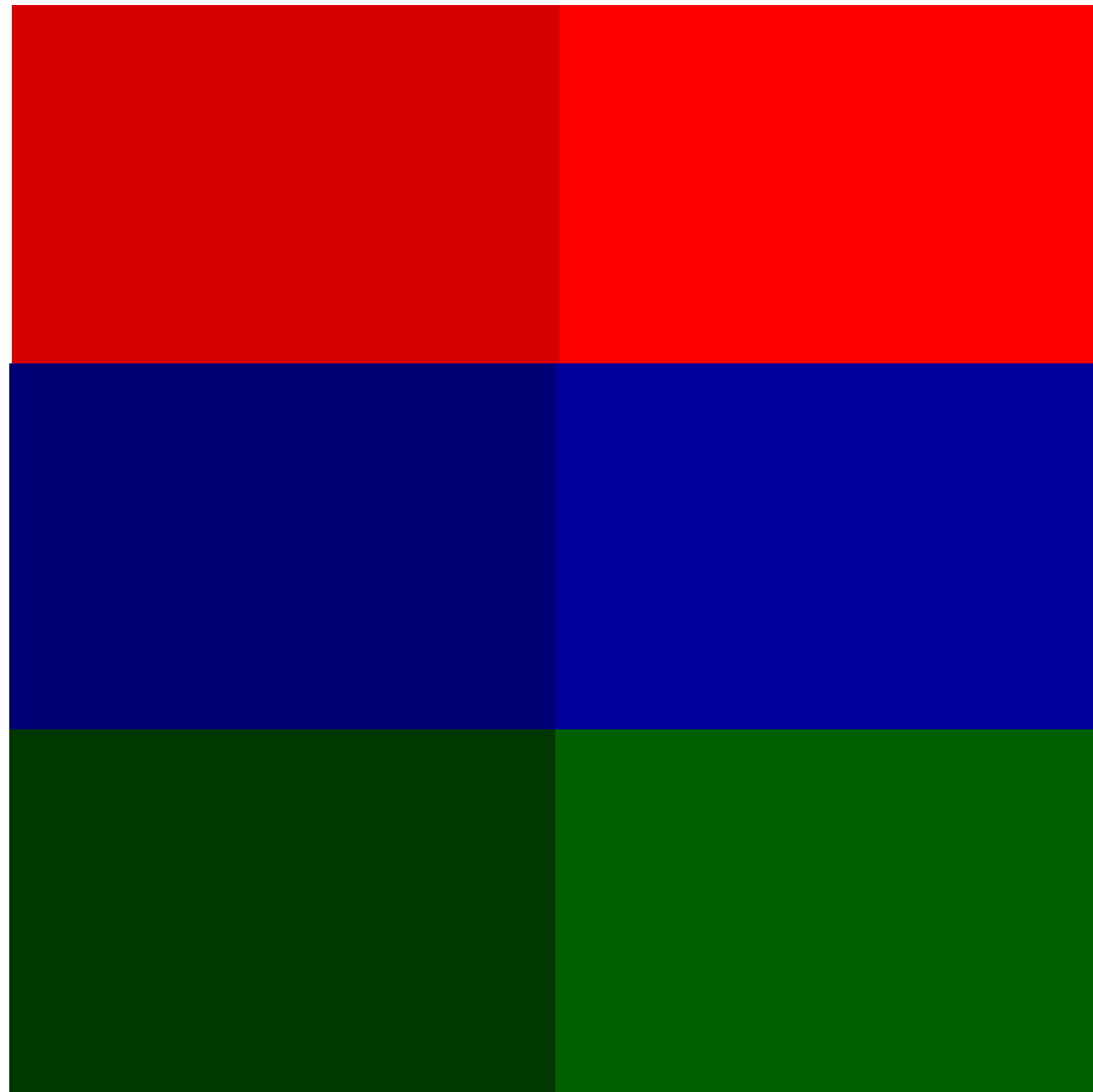
Time to Cheat!

Find shots with colors that look like what you're searching for.



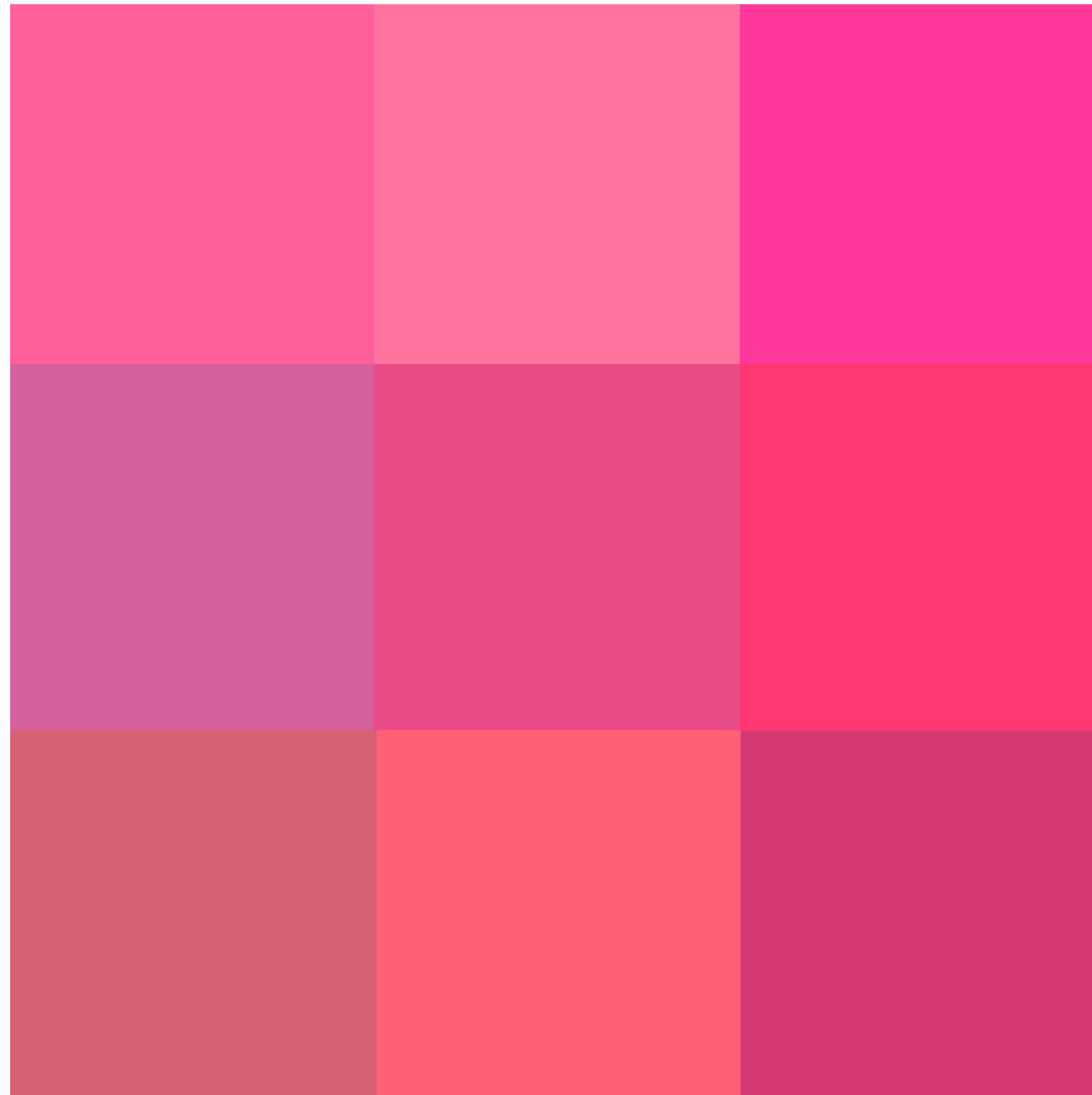




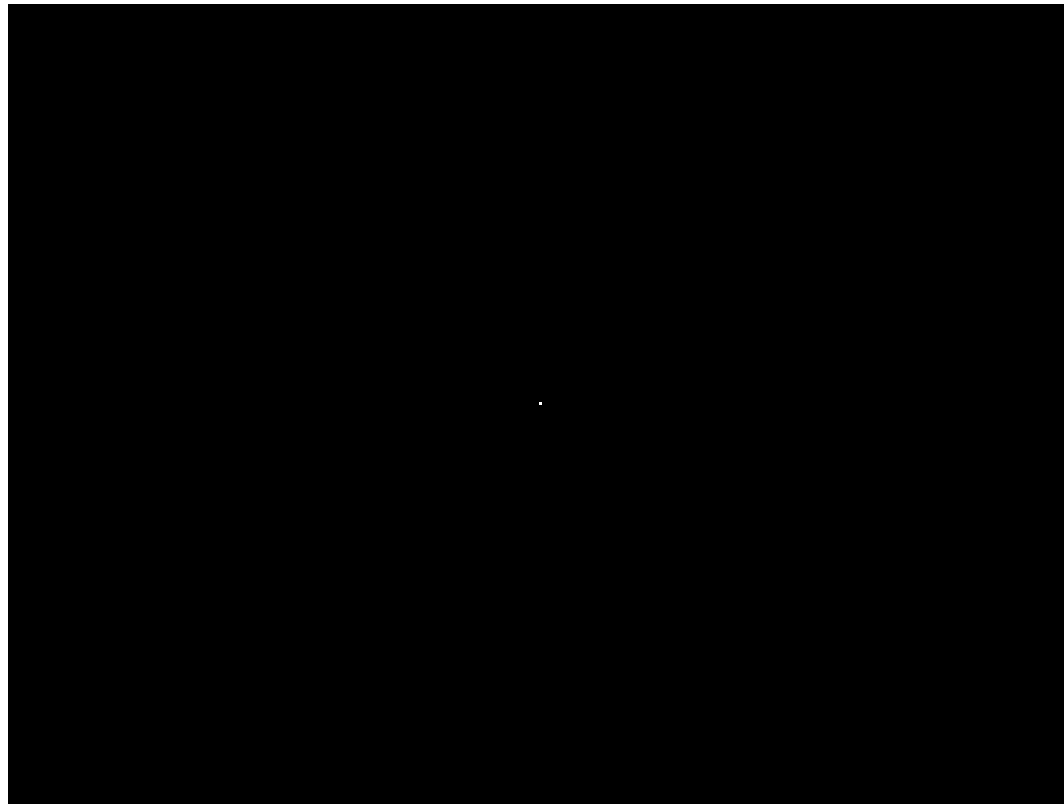


Searching

```
WHERE ((red BETWEEN min AND max) AND  
      (green BETWEEN min AND max) AND  
      (blue BETWEEN min AND max))
```



Problem



Minimum Percent

Where $\text{percent} > \text{Minimum_Percent}$

Summary

- Quantize image to 8 colors



Summary

- Quantize image to 8 colors
- Look for colors that look like the search



Summary

- Quantize image to 8 colors
- Look for a color “near” the search
- Look for shots with a decent percentage of color



Thank You!

